Thesis Update #3

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Current Goal

Validate literature-based NIR wavelength ranges for glucose detection using a reflectance and transmission mode setup for comparing wavelength-specific glucose sensitivity and SNR

Research Paper #1:
Yang et al., 2018
Determination of NIR informative wavebands for transmission non-invasive blood glucose measurement using a Fourier transform spectrometer [1]

Research Paper #1

Data Collection Protocol:

- Glucose solution experiment
  - Obtained NIR transmission spectra for glucose solution
  - Pure water + glucose solution heated to match body temperature
  - Varying concentrations tested to determine glucose sensitive NIR wavebands
  - Glucose sensitive wavebands used for NIR fingertip tests
- Human fingertip experiment
  - 7 male subjects completed OGTT and fingertip NIR transmission measurements
  - OGTT induced changing BGL over time
  - Fingertip NIR spectra and invasive BGL measurements were repeated and averaged
  - Measure correlation coefficient, R, between change in transmitted light intensity and invasively measured BGL

Research Paper #1

Results:

- 950nm and 1280nm provided R>0.72 and usable SNR for transmission mode
- 1650nm showed glucose sensitivity in solution but poor fingertip SNR
- Seven male volunteers tested so the small and non-diverse dataset limits generalizability
- No shielding protocol described for reducing external NIR interference
- Results may be different for reflectance mode setup

Research Paper #2:
Ge et al., 2024
“Evaluation and Validation on Sensitivity of Near-Infrared Diffuse Reflectance in Non-Invasive Human Blood Glucose Measurement”[2]

Research Paper #2

Overview

- Glucose in tissue
  - Changes NIR absorption
  - Changes refractive index matching→changes scattering
- Reflected intensity, I_R, depends on both absorption and back scattering
- $Apparent\ absorbance\ (A) = -\log(\frac{I_R}{I_0})$
- Normalize w.r.t reference intensity, I_0, which depends on setup
- Sensitivity of apparent absorbance w.r.t glucose concentration depends on wavelength
- At 1550nm, apparent absorbance sensitivity -0.00107 a.u./ (mmol/L), the largest compared to other tested wavelengths (1050, 1219, 1314, 1380, and 1609 nm)
- 1550nm good candidate wavelength for reflectance mode since reflected intensity is the most sensitive at this wavelength
- Results of previous studies have differed due to significant differences between living and synthetic tissue

Research Paper #2

Data Collection Protocol

- Two source-detectors separations at 1.7mm and 2.0mm
  - Samples photons travelling different path lengths
  - Removes common-mode interference between detectors
- Each detector measures apparent absorbance
- $Differential\ absorbance = A_2 - A_1$
- Larger data set, 25 people
- Stable forearm posture to maintain similar photon transmission path
- Light guided by optical fibers
- Detector made of photosensitive InGaAs

Research Paper #3:
Rajeswari 2024
Development of sensor system and data analytic framework for non-invasive blood glucose prediction[3]

Research Paper #3

Overview:

- Focused on developing an accurate, portable, and low-cost NIR reflectance glucose sensor
- Used preprocessing and ridge regression ML model to estimate blood glucose
- Linear vs. Ridge Regression
  - Linear: $BGL = w_T x + b$
    - Where w and x are the weight and NIR feature vector, respectively
    - Weights can become too large if data is noisy
    - Risk of overfitting so model may fail on new people
  - Ridge: $BGL = w_T x + b$
    - Same equation, but a penalty added to large weights
- Ridge regression performed best under subject-dependent factors such as skin color variation, finger pressure

Research Paper #3

Test Setup:

- Used
  - 940 nm NIR emitter based on literature and penetration depth
  - 900-1700 nm detector, 0.9A/W responsivity
  - Convert detector current to voltage using TIA, $V_{sense} = I_{sense} R_{amp}$
  - Finger and wrist sensor placement
  - Normalization to remove dark current effect for photodiodes
  - Optical shielding: PCB covered in black body with transparent window
  - Consistent sensor pressure for good SNR and contact geometry

Plan

- LTSpice Simulation [3]
  - Photodiode current
  - TIA amplifier
  - Low-pass filter
- Testing
  - Prepare multiple glucose concentrations
  - Measure background noise (LED off and no ambient source) [3]
  - Measure reflected intensity in pure water, I_0 [2]
  - Validate from research
    - Candidate wavelengths 940/950, 1280/1300, 1550 nm [1][2][3]
    - Non-candidate wavelengths
    - Sensor placements (finger and wrist) [3]
    - Source-detector separation (1.7mm and 2.0mm for reflectance mode) [2]
  - Determine apparent absorbance sensitivity for best wavelength and sensor configuration [2][3]
  - Results for glucose water samples may not match with in vivo tests [2]

Plan

- Setup [3]
  - NIR diodes
  - Photodiode
  - TIA Amplifier
  - Narrow bandpass filter: Only allows specific wavelength to pass
  - ADC

In-Progress Research

Research Paper #4:
A. Hina and W. Saadeh., 2020
A noninvasive glucose monitoring SoC based on single wavelength photoplethysmography

Research Paper #4

Overview:

- 940 nm NIR PPG glucose-monitoring SoC
- Transmission mode
  - Photodiode converts optical light to current
  - Current converted to voltage with TIA
  - Voltage signal is filtered and then digitized
  - PPG features extracted
  - ML regression: Fine Gaussian Support Vector Regression
  - BGL estimation
- Total Soc power of 1.62 mW

Research Paper #4

Test Setup:

- LED Driver
  - 0.5V → 1.2V using boost converter
  - Pulsed LED to reduce power
- PPG Readout Circuit
  - Integration-mode switch-capacitor TIA with $1\ M\Omega$ gain
  - Replaaces need for actual 1 $M\Omega$ resistor
- 10 Hz low-pass filter
- DC current rejection up to 20 μA
- ADC

Research Paper #5:
Srichan et al., 2022
Non-invasively accuracy enhanced blood glucose sensor using shallow dense neural networks with NIR monitoring and medical features[4]

Subject Independent Glucose Estimation

Results:

- Used PMF to more accurately estimate BGL levels
- Can be used to assist BGL while estimation still remains subject independent

References

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W. Yang, N. Liao, H. Cheng, Y. Li, X. Bai, and C. Deng, “Determination of NIR informative wavebands for transmission non-invasive blood glucose measurement using a Fourier transform spectrometer,” *AIP Advances*, vol. 8, no. 3, Art. no. 035216, 2018, doi: 10.1063/1.5017169.
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[2] Ge et al., 2024
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[4] A. Hina and W. Saadeh, “A noninvasive glucose monitoring SoC based on single wavelength photoplethysmography,” *IEEE Transactions on Biomedical Circuits and Systems*, vol. 14, no. 3, pp. 504–515, Jun. 2020, doi: 10.1109/TBCAS.2020.2979514. Available: https://ieeexplore.ieee.org/abstract/document/9028112

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C. Srichan, W. Srichan, P. Danvirutai, C. Ritsongmuang, A. Sharma, and S. Anutrakulchai, “Non-invasively accuracy enhanced blood glucose sensor using shallow dense neural networks with NIR monitoring and medical features,” *Scientific Reports*, vol. 12, Art. no. 1769, 2022, doi: 10.1038/s41598-022-05570-8.
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